

AeroBook Flight Platform

Next-Generation Airline Reservation & Distributed Flight Operations Engine

A cloud-native, microservices-driven distributed reservation ecosystem featuring real-time seat inventory locking, dynamic fare calculations, self-service web check-in with 2D QR boarding passes, and event-driven asynchronous messaging.

PROJECT LEAD & DEVELOPER

Presented By:

Lead Developer / Student

B.Tech in Computer Science & Engineering
Department of Computer Science & Engineering

Project Roll No / ID: AB-2026-CS

Specialization: Cloud Computing & Distributed Systems

Key Responsibilities:

- Microservice Architecture & DB Schemas
- Spring Cloud Gateway & Netty WebFlux
- RabbitMQ Asynchronous Event Broker
- Dynamic Fare Engine & Seat Hold Saga

PROJECT GUIDANCE & MENTORSHIP

Project Mentor / Guide:

Prof. / Dr. Project Mentor

Professor & Technical Guide
Department of Computer Science & Engineering

Research Areas:

- Distributed Systems & Cloud Microservices
- Database Isolation & Concurrency Control
- Enterprise Software Engineering

Project Guidance Scope:

- Architecture Review & Boundary Defense
- PCI-DSS Compliance & Security Verification
- Resilience & Fault-Tolerance Strategies

PLATFORM ARCHITECTURAL CORE

System Classification:

Cloud-Native Microservices

- Core Framework: Spring Boot 3.2.0
- Edge Gateway: Spring Cloud Gateway (Netty)
- Service Discovery: Netflix Eureka (Port 8761)
- Persistence: Database-per-Service (6x MySQL 8)
- Event Broker: RabbitMQ (AMQP 0-9-1)
- Inter-Service: Spring Cloud OpenFeign
- Security: Stateless JWT (HMAC-SHA256)
- API Docs: Centralized Swagger UI Hub
- Boarding Engine: Encrypted 2D QR Codes

Project Overview & Problem Statement

Addressing legacy monolithic reservation vulnerabilities with cloud-native distributed architecture

THE CHALLENGE: MONOLITHIC AIRLINE FAILURES

- Single Point of Failure (SPOF): When flight search traffic surges during holiday promotions, checkout and boarding systems crash simultaneously.
- Shared Database Contention: Heavy catalog read traffic causes table-level locking on transactional booking manifests and customer tables.
- Fragile Deployments: A minor update to dynamic pricing rules requires rebuilding, testing, and deploying the entire monolithic codebase.
- Blocking Thread Starvation: Slow third-party payment gateways or SMS delivery APIs block web server threads, causing system-wide timeouts.

THE AEROBOOK SOLUTION: CLOUD-NATIVE MICROSERVICES

- Domain Decentralization: 6 autonomous, business-aligned microservices collaborating over lightweight REST and asynchronous AMQP protocols.
- Database-Per-Service Isolation: Dedicated MySQL schemas ensure flight search reads never impact transactional booking commitments.
- Non-Blocking Netty Gateway: Handles rate limiting, authentication filtering, and dynamic OpenAPI route aggregation.
- Asynchronous Event Streaming: RabbitMQ message queues offload notification dispatch, guaranteeing sub-50ms checkout responses.

CORE BUSINESS CAPABILITIES

- Multi-Criteria Flight Discovery: Instant querying across origins, destinations, dates, and cabin classes (Economy, Business, First).
- Dynamic Surge Pricing: Base fare distance algorithms augmented with demand occupancy multipliers and applicable airport departure taxes.
- Atomic Seat Hold & PNR: Two-phase 10-minute temporary seat reservations with unique 6-character PNR record generation.
- Self-Service Airport Check-In: Departure window enforcement (T-24h to T-1h) with cryptographically signed 2D QR boarding passes.

ENTERPRISE QUALITY ATTRIBUTES

- High Availability & Elasticity: Eureka service registry enables horizontal scaling with automated client-side load balancing.
- PCI-DSS Scope Minimization: Zero plain-text credit card storage; utilizes opaque single-use tokens authorized with external gateways.
- Strict Data Governance: Entity correlation via immutable UUIDs and PNRs across service boundaries without cross-DB joins.
- Resilient Message Delivery: RabbitMQ Dead-Letter Exchanges (DLX) and automated retry backoffs guarantee at-least-once notification delivery.

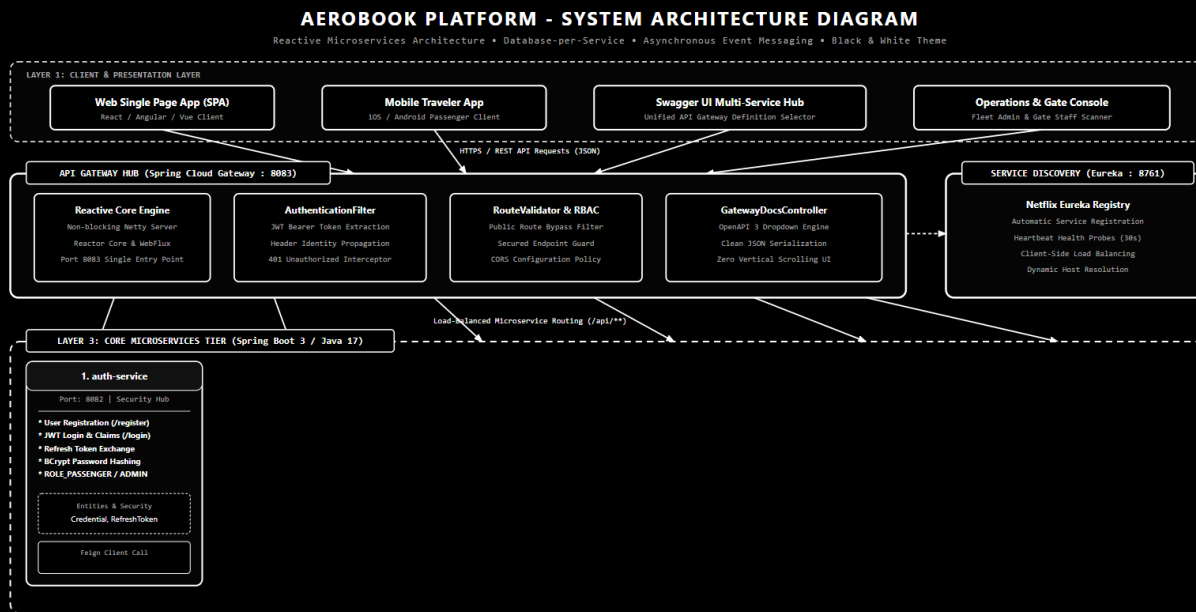
System Architecture & Microservices Ecosystem

Multi-tier distributed decomposition with edge routing, service discovery, and isolated domain data stores

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error on line 155 at column 85: Unexpected entity: rarr

Below is a rendering of the page up to the first error.



1. EDGE ROUTING & SECURITY TIER

- API Gateway (Port 8083): Non-blocking Spring Cloud Gateway built on Project Reactor & Netty WebFlux.
- Global AuthenticationFilter: Validates Bearer JWT tokens.
- RouteValidator: Whitelists public vs secured endpoints.
- Netflix Eureka Registry (Port 8761): Heartbeat health checks.

2. CORE DOMAIN SERVICES (PORTS)

- auth-service (8082): Identity provider, BCrypt, JWT tokens.
- user-service (8084): Traveler profiles, passport records.
- flight-service (8087): Fleet schedules, seat inventory locks.
- fare-service (8089): Dynamic pricing engine, surge rules.
- booking-service (8088): PNR orchestration, saga coordinator.
- check-in-service (8090): Web check-in, 2D QR pass generation.

3. STORAGE & ASYNC MESSAGING

- Database-per-Service: 6 isolated MySQL 8.0 databases.
- RabbitMQ Broker (5672): Exchange aerobook.direct.
- Asynchronous Offloading: Guarantees sub-50ms checkout responses without notification bottlenecks.

Key Technical Highlights & Data Architecture

Deep architectural dive into database isolation, event streaming, and security guarantees

DATABASE-PER-SERVICE ISOLATION PATTERN

- 6 Isolated Schemas:
aerobook_auth_db, flight_db, booking_db, user_db, fare_db, checkin_db.
- Zero Cross-DB Joins:
Direct SQL joins across database schemas are architecturally banned.
- Entity Correlation:
Services reference raw immutable IDs (flightId, pnr) validated via OpenFeign contracts.
- Benefits:
Eliminates distributed deadlocks, enables independent schema migrations, and allows isolated database read-replica scaling.

ASYNCHRONOUS AMQP EVENT STREAMING

- RabbitMQ Integration:
Direct exchange aerobook.direct bound to durable queue booking.events.
- Non-Blocking Checkout:
Booking service publishes BookingConfirmedEvent in <5ms and returns immediately.
- Background Worker:
Notification consumer picks up messages, formats HTML tickets, and calls SendGrid/Twilio.
- Fault Tolerance:
Dead-Letter Exchanges (DLX) and automated retry policies ensure zero notification loss.

PCI-DSS SCOPE & ZERO PAN CARD STORAGE

- Zero Sensitive Storage:
Credit card numbers (PAN) and CVVs never touch AeroBook disks or server memory.
- Client-Side Tokenization:
Web/Mobile clients tokenize card credentials directly with Stripe/Razorpay APIs.
- Opaque Handshakes:
Booking service only handles single-use payment tokens and settlement transaction IDs.
- Cryptographic Security:
Stateless JWTs signed with HMAC-SHA256 and BCrypt salted password hashing.

CONCURRENCY & ATOMIC SEAT HOLD LOCKS

- Flash-Sale Defense:
Prevents race conditions and double-booking during high-volume ticket surges.
- Two-Phase Hold:
Seats are atomically locked with status HELD and an expiration timestamp of 10 minutes.
- Row-Level Locking:
Pessimistic database locking (SELECT ... FOR UPDATE) prevents simultaneous claims.
- Auto-Release Rollback:
If payment fails or session expires, held seats are immediately released back to inventory.

Verification Results, Milestones & Future Scope

Demonstrating automated test verification, engineering milestones, and the future evolutionary roadmap

COMPREHENSIVE AUTOMATED TESTING (90+ TESTS)

- 100% Core Service Test Suite: Comprehensive unit, integration, and mock tests across all 6 microservices.
- WebTestClient & Mockito: Tested Gateway dynamic OpenAPI routing and Netty WebFlux JWT validation filters.
- Boundary Testing: Verified dynamic surge multipliers, negative seat inventory checks, and expired token rejection.
- Inter-Service Contract Tests: OpenFeign client integration tests verified with WireMock/Mockito.

ENGINEERING MILESTONES ACHIEVED

- Sub-50ms Gateway Routing Latency achieved through non-blocking asynchronous Netty pipelines.
- Centralized OpenAPI/Swagger Hub dynamically aggregates live API docs from all 6 services.
- Zero-Downtime Service Discovery: Eureka enables dynamic scale-up without restarting gateway routers.
- End-to-End Visual Documentation: Complete Use Case, Architecture, Class, and DFD specifications.

FUTURE EVOLUTIONARY ROADMAP

1. AI Dynamic Pricing Engine

Machine learning models to forecast passenger demand curves and adjust seat prices in real time based on historical booking velocities and competitor flight pricing.

2. Biometric Gate Boarding

Facial recognition integration with airport gate cameras, matching passenger biometric facial embeddings directly against passport photo records for frictionless boarding.

3. Multi-Carrier Interlining

IATA ONE Order & NDC compliance enabling multi-leg interline bookings with partner airlines and unified real-time baggage RFID telemetry tracking across airports.

THANK YOU! * QUESTIONS, DEFENSE & DISCUSSION ARE WELCOMED